ALERT meeting - Kalman Filter updates

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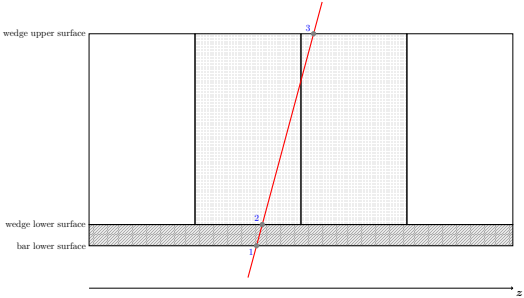
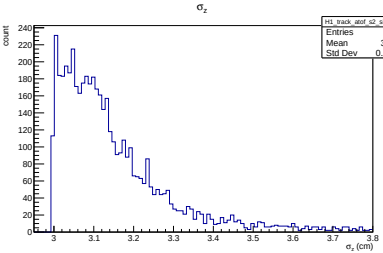
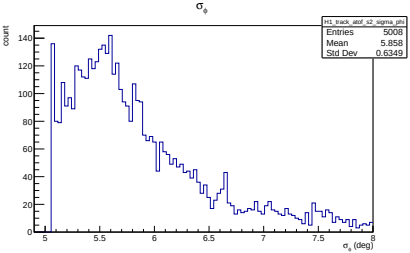
Updates (ongoing work)

- Still working on adding the ATOF hit to the Kalman Filter
  - Use the **Propagator** of the KF to project the AHDC track to the ATOF
  - Use the (x, y, z) coordinates on the surface 1, 2, 3 to predict the bar/wedge
  - Some limitations: AHDC-ATOF alignment, resolution on z and on ϕ



Updates (ongoing work)

- Some statistics
  - **nb. predicted wedge ai based / nb. tracks =** 29.76%
  - nb. hit S2 / ai prediction $= 12.52\%$ (exact match)
  - nb. hit S3 / ai prediction $= 10.51\%$ (exact match)
  - nb. hit S2 or S3 / ai prediction $= 18.95\%$ (exact match)
  - nb. hit S2 / ai prediction $= 70.16\%$ (looked also at neighbors)
  - nb. predicted wedge proj based / nb. tracks $= 37.59\%$ (the wedge in S2 or S3 exists in ATOF::hits)
  - nb. predicted wedge proj based / nb. tracks $= 43.49\%$ (the wedge in S2 or its neighbors exists in ATOF::hits)

Updates (ongoing work)

- Resolutions on z and on ϕ for the hit on the surface S2 : consistent with the ± 1 indetermination on the layer or the wedge ids